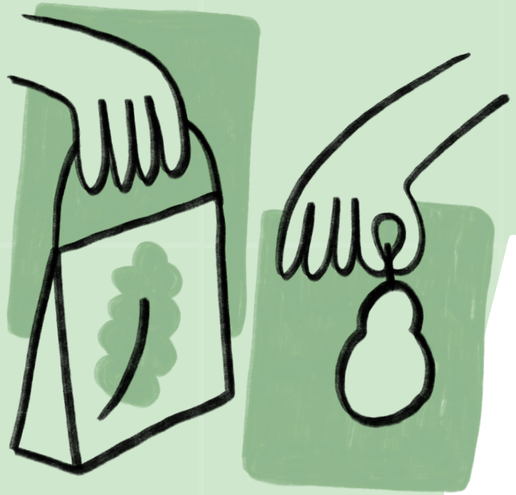

# GROCERY MARKET ANALYTICS

## Quick Overview

# Project Overview

- Smart Grocery Analytics is an interactive data analysis application developed using R and Shiny, designed to explore and analyze grocery store transaction data.

- The project aims to transform raw grocery datasets into meaningful insights through data cleaning, visualization, clustering, and association rule mining.

- This project is considered a Data Analytics project as it includes data cleaning, visualization, and advanced analytical techniques such as clustering and association rule mining to extract meaningful insights and support decision-making.

# Project Objectives

- Clean and preprocess raw data.
- Visualize key business insights.
- Analyze customer behavior.
- Segment customers using clustering.
- Discover relationships between purchased items.

# features

**1. Data Input & Cleaning**

- Users can upload a dataset using a file path
- Automatically detects:
  - Duplicate records
  - Missing values
  - Outliers
  -
- Cleans data for accurate analysis

**2. Interactive Dashboard**

The dashboard provides multiple visualizations:

- Comparison between cash and credit payments
- Total spending grouped by age
- Total spending by city (sorted descending)
- Distribution of total spending

**3. Interactive Dashboard**

- Uses K-Means Clustering
- Groups customers based on:
  - Age
  - Total spending
- Displays:
  - Customer Name
  - Age
  - Total Spending
  - Cluster Assignment

**4. Association Rule Mining**

- Uses Apriori Algorithm
- Discovers relationships between purchased items
- User controls:
  - Minimum Support
  - Minimum Confidence
  - Minimum Rule Length

# Technologies Used

- R Programming Language
- Shiny (for interactive UI)
- dplyr (data manipulation)
- plotly / base plots (visualization)
- arules (association rules)
- RColorBrewer (color palettes)

# Inputs

- Dataset (CSV file path)
- Number of clusters (2–4)
- Minimum support (0.001–1)
- Minimum confidence (0.001–1)
- Minimum rule length (2–10)

# Outputs

- Cleaned dataset summary
- Visual dashboard
- Customer clusters table
- Association rules between items

# Key Insights

**This project helps in:**

- Understanding customer spending behavior
- Identifying high-value customer segments
- Discovering purchasing patterns
- Supporting data-driven decision making in retail